EM field

$$\begin{cases} \nabla \times E = -\dfrac{\partial B}{\partial t} & (1) \\ \nabla \times B = \mu_0\left(\vec{j} + \varepsilon_0 \dfrac{\partial \vec{E}}{\partial t}\right) & (2) \\ \nabla \cdot E = \dfrac{\rho}{\varepsilon_0} & (3) \\ \nabla \cdot B = 0 & (4)\ \mu_0 \varepsilon_0 c^2 = 1 \end{cases}$$

$$\vec{\nabla} \cdot \vec{\nabla} \times \vec{B} = 0 = \vec{\nabla} \cdot \vec{j} + \varepsilon_0 \frac{\partial}{\partial t} \underbrace{\vec{\nabla} \cdot \vec{E}}_{\frac{\rho}{\varepsilon_0}} = 0$$

$$\Rightarrow \frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot \vec{j} = 0$$

Vector and scalar potentials:

$$\vec{B} = \vec{\nabla} \times \vec{A}, \quad \vec{E} = -\vec{\nabla}\varphi - \frac{\partial \vec{A}}{\partial t}$$

Note that (1) and (4) are satisfied automatically.

$$\vec{\nabla} \cdot \vec{E} = -\nabla^2 \varphi - \frac{\partial}{\partial t}\left(\vec{\nabla} \cdot \vec{A}\right) = \frac{\rho}{\varepsilon_0} \qquad (5)$$

$$\underbrace{\vec{\nabla} \times \vec{\nabla} \times \vec{A}}_{\vec{\nabla}(\vec{\nabla} \cdot \vec{A}) - \nabla^2 \vec{A}} - \frac{1}{c^2} \frac{\partial}{\partial t}\left(-\vec{\nabla}\varphi - \frac{\partial \vec{A}}{\partial t}\right) = +\mu_0 \vec{j}$$

$$\nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} - \vec{\nabla}\left(\vec{\nabla} \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t}\right) = -\mu_0 \vec{j} \qquad (6)$$

* Thus, 4 equs for $\vec{E}, \vec{B}$, but only 2 for $\vec{A}, \varphi$.

Gauge invariance : φ, $\vec{A}$ are <u>not unique</u>

$$\vec{A} \to \vec{A}' = \vec{A} + \vec{\nabla}\chi \quad , \quad \varphi \to \varphi' = \varphi - \frac{\partial \chi}{\partial t}$$

$$\vec{B} = \vec{\nabla} \times \vec{A} \to \vec{\nabla} \times \vec{A}' = \vec{\nabla} \times \vec{A} + \vec{\nabla} \times (\vec{\nabla}\chi) = \vec{\nabla} \times \vec{A} \text{ as before.}$$

$$\vec{E} = -\vec{\nabla}\varphi - \frac{\partial \vec{A}}{\partial t} = -\vec{\nabla}\left(\varphi - \frac{\partial \chi}{\partial t}\right) - \frac{\partial}{\partial t}\left(\vec{A} + \vec{\nabla}\chi\right) =$$

$$= -\vec{\nabla}\varphi - \frac{\partial \vec{A}}{\partial t} \quad \text{too.}$$

$\Rightarrow$ can choose $\chi(\vec{r},t) \Rightarrow$ every choice represents one of allowed <u>gauges</u>

1) Lorenz gauge $\quad \vec{\nabla} \cdot \vec{A} + \frac{1}{c^2}\frac{\partial \varphi}{\partial t} = 0$

$$\Downarrow$$

$$\vec{\nabla} \cdot \left(\vec{A} + \vec{\nabla}\chi\right) + \frac{1}{c^2}\frac{\partial}{\partial t}\left(\varphi - \frac{\partial \chi}{\partial t}\right) = 0$$

$$\Rightarrow \nabla^2 \chi - \frac{1}{c^2}\frac{\partial^2 \chi}{\partial t} = -\vec{\nabla} \cdot \vec{A} - \frac{1}{c^2}\frac{\partial \varphi}{\partial t}$$

$\Leftarrow$ eqn. on $\chi(\vec{r},t)$ for given $\vec{A}$, φ.

Lorenz gauge is Lorentz covariant:

$$(5) \to \quad -\nabla^2 \varphi - \frac{\partial}{\partial t}\left(-\frac{1}{c^2}\frac{\partial \varphi}{\partial t}\right) = \frac{\rho}{\varepsilon_0}$$

$$\frac{1}{c^2}\frac{\partial^2 \varphi}{\partial t^2} - \nabla^2 \varphi = \frac{\rho}{\varepsilon_0} \quad (5')$$

$$(6) \to \quad \frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} - \nabla^2 \vec{A} = \mu_0 \vec{j} \quad (6')$$

$$\Box \varphi = \frac{\rho}{\varepsilon_0}$$

$$\Box \vec{A} = \mu_0 \vec{j}$$

$\Rightarrow$ 4-vector $J^\alpha = (c\rho, \vec{J})$, 4-potential $A^\alpha = \left(\frac{\varphi}{c}, \vec{A}\right)$

$$\boxed{\partial_\alpha \partial^\alpha A^\beta = \mu_0 J^\beta}$$

$$\partial_\alpha u = \left(\frac{1}{c}\frac{\partial u}{\partial t}, \vec{\nabla} \cdot \vec{u}\right)$$

$$\partial^\alpha u = \left(\frac{1}{c}\frac{\partial u}{\partial t}, -\vec{\nabla} \cdot \vec{u}\right)$$

2) Coulomb (or, radiation) gauge $\vec{\nabla}\cdot\vec{A}=0$

$$\Rightarrow \vec{\nabla}\cdot\vec{A}'=0 \Rightarrow \vec{\nabla}\cdot\left(\vec{A}+\vec{\nabla}\chi\right)=0 \Rightarrow$$

choose χ such that $\nabla^2\chi=-\vec{\nabla}\cdot\vec{A}$, then

$$\vec{\nabla}\cdot\vec{A}'=0$$

Then (5) $\rightarrow$ $\nabla^2\varphi=-\frac{\rho}{\varepsilon_0}$ (5'')

(6) $\rightarrow$ $\nabla^2\vec{A}-\frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2}=-\mu_0\vec{j}+\frac{1}{c^2}\frac{\partial}{\partial t}\vec{\nabla}\varphi$ (6'')

In this gauge ~~static~~ scalar potential φ is not retarded (determined by instantaneous charge distribution $\rho(\vec{r},t)$)

$\Rightarrow$ radiation field is determined by $\vec{A}(\vec{r},t)$ alone.

Energy $\left.\begin{aligned} U &= \frac{1}{2}\varepsilon_0\left(\vec{E}^2+c^2\vec{B}^2\right) \\ \vec{S} &= \frac{1}{\mu_0}\vec{E}\times\vec{B}\end{aligned}\right\}$ $\vec{\nabla}\cdot\vec{S}+\frac{\partial U}{\partial t}=0$

let's now analyze EM field in vacuum (no charges).
In Coulomb gauge (6'') reads

$$\nabla^2 \vec{A} - \frac{1}{c^2}\frac{\partial^2 \vec{A}}{\partial t^2} = 0 \quad \leftarrow \text{simple wave eqn!}$$

$$\Rightarrow \vec{A}(\vec{r},t) = \sum_{\vec{k};\alpha=1,2} \vec{e}_\alpha\, a_k\, e^{i(\vec{k}\cdot\vec{r}-\omega_k t)} + \vec{e}_\alpha\, a_k^*\, e^{-i(\vec{k}\cdot\vec{r}-\omega t)}$$

$\vec{e}_\alpha$ = (complex) polarization vectors

From $\vec{\nabla}\cdot\vec{A} = 0$ follows $\vec{e}\cdot\vec{k} = 0 \Rightarrow \alpha = 1, 2$

(two transverse polarizations)